

Dabin Park

Ph.D. Student, School of Mathematics, Kyungpook National University

Research Assistant, Mathematics of Tomography Lab

Daegu, Republic of Korea · Advisor: Prof. Sunghwan Moon

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RESEARCH INTERESTS

Inverse problems; integral geometry and tomography (photoacoustic tomography, Radon-type transforms); operator learning and scientific machine learning for partial differential equations; finite element methods.

EDUCATION

Kyungpook National University — Ph.D. student in Mathematics 2023.09 – present

Kyungpook National University — M.S. in Mathematics 2022.03 – 2023.08

- GPA 4.30 / 4.30

Kyungpook National University — B.S. in Mathematics, *Magna Cum Laude* 2018.03 – 2022.02

- GPA 3.81 / 4.30 (major 4.14 / 4.30)

RESEARCH EXPERIENCE

Mathematics of Tomography Lab, Kyungpook National University — Research Assistant 2022.03 – present

- Advisor: Prof. Sunghwan Moon.
- Inverse problems for the wave and fractional wave equations; deep learning for photoacoustic tomography and operator learning for parametric PDEs.

Florida State University, Tallahassee, FL, USA — Visiting Researcher 2024.04 – 2025.07

- BK21 FOUR Graduate Student Overseas Training Program: 12-month fellowship, extended by a further 3 months at own expense. Host: Prof. Sanghyun Lee.
- Inverse and multiscale problems for physical equations; machine learning and finite element methods for inverse elliptic PDEs.

PUBLICATIONS

Peer-reviewed journal articles

1. **D. Park**, S. Lee, S. Moon. *A machine learning and finite element framework for inverse elliptic PDEs via Dirichlet-to-Neumann mapping*. Journal of Computational and Applied Mathematics, **486** (2026), 117679. [doi:10.1016/j.cam.2026.117679](https://doi.org/10.1016/j.cam.2026.117679)
2. G. Hwang, G. Jeon, S. Moon, **D. Park**. *Implicit learning to determine variable sound speed and the reconstruction operator in photoacoustic tomography*. Applied Mathematics in Science and Engineering, **34**(1) (2026), 2661458. [doi:10.1080/27690911.2026.2661458](https://doi.org/10.1080/27690911.2026.2661458)
3. **D. Park**, S. Moon. *Reconstruction of an initial function from the solutions of the fractional wave equation on the light cone trace*. Mathematical Methods in the Applied Sciences, **47**(12) (2024), 9469–9475. [doi:10.1002/mma.10078](https://doi.org/10.1002/mma.10078)

Refereed conference proceedings

4. F. Saeed, **D. Park**, S. S. A. Shah, J. Lee, A. Rehman, J.-M. Kang, I.-M. Kim, S. Yun, S. Moon. *Secure and privacy-preserving intrusion detection in wireless sensor networks via perturbation adversarial training*. Proceedings of the 41st ACM/SIGAPP Symposium on Applied Computing (SAC '26), Thessaloniki, Greece (2026), 2081–2087. [doi:10.1145/3748522.3779960](https://doi.org/10.1145/3748522.3779960)

In preparation

5. **D. Park**, S. Lee, S. Moon. *T-FEONet: A Transformer-based finite element operator network for time-dependent parametric PDEs*.

AWARDS

Top Prize , Graduate Student Paper Presentation Contest, Fall Conference of the Korean Data and Information Science Society	2023.11
Participation Prize , Annual Meeting and Research Presentation of the Youngnam Mathematical Society	2023.07
Award for Presentation Excellence , 15th Graduate Student Workshop on Mathematics	2022.07
Magna Cum Laude on B.S. graduation, Kyungpook National University	2022.02

CONFERENCE PRESENTATIONS

Secure and privacy-preserving intrusion detection in wireless sensor networks via perturbation adversarial training

- **Oral**. 41st ACM/SIGAPP Symposium on Applied Computing (SAC '26), Grand Hotel Palace, Thessaloniki, Greece, 2026.03.

A machine learning and finite element framework for inverse elliptic PDEs via Dirichlet-to-Neumann mapping

- **Oral (contributed student talk)**. 2025 Southeast Applied and Computational Math Student Workshop, Auburn University, Auburn, AL, USA, 2025.04.06.

Implicit learning for the variable sound speed and the reconstruction operator in photoacoustic tomography

- **Oral**. Fall Conference of the Korean Data & Information Science Society, Daegu University, 2023.11.03–11.04. Top Prize

Reconstruction of the initial functions from the solutions of the fractional wave equation on the light cone trace

- **Poster**. Annual Meeting of the Youngnam Mathematical Society, Kyungpook National University, Daegu, 2023.07.14–07.16. Participation Prize
- **Oral (in English)**. 15th Graduate Student Workshop, online, 2022.07.26–07.28. *Award for Presentation Excellence*
- **Oral**. Youngnam Mathematical Society, Dong-A University, Busan, 2022.06.23–06.24.
- **Poster**. KSIAM Spring Conference, Institute for Basic Science, 2022.05.27–05.29.
- **Oral**. Harmonic Analysis Workshop, Sungkyunkwan University, online, 2022.02.16–02.18.

SERVICE

Organizing Committee, Workshop for Young Mathematicians in Korea (WYMK) 2026

CONFERENCES AND WORKSHOPS ATTENDED

ICIAM 2023, Waseda University, Tokyo, Japan	2023.08.20 – 08.25
KMS Spring Meeting, Daejeon Convention Center	2023.04.27 – 04.29
Winter AI Tutorial 2023 — Graph Neural Networks and Generative Models	2023.02.21 – 02.22
KSIAM Fall Conference, Sonocalm Jeju	2022.11.24 – 11.27
KSIAM Fall Conference, BEXCO, Busan	2021.12.02 – 12.05

COMPETITIONS

Nongnet Agricultural Price Fluctuation Forecasting AI Competition 2022.09

- Predicting the rate of change of agricultural prices against a reference date.
- **18th place** out of 69 teams. [Competition page](#)

SKILLS

- **Scientific computing:** FEniCS / legacy dolfin, finite element methods
- **Machine learning:** PyTorch (CUDA), operator learning, Transformer architectures
- **Languages and tools:** Python, Julia, MATLAB, $\text{\LaTeX}$, Git, Docker / Dev Containers, Linux (WSL2)